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Data-driven state of health and state of safety estimation for alternative battery chemistries — A comparative review focusing on sodium-ion and LFP lithium-ion batteries

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ABSTRACT

This paper presents a comprehensive survey on data-driven online estimation of state of health (SoH) for alternative battery chemistries for maritime applications, with a particular focus on LFP lithium-ion and sodium-ion types of batteries. In addition, the emerging concept of state of safety (SoS), a critical yet underexplored metric for maritime battery systems, is explored. Building on previous work on nickel–manganese–cobalt (NMC) lithium-ion batteries, this study evaluates the applicability of existing SoH estimation methodologies to alternative chemistries. The findings suggest that similar data-driven approaches, including empirical and semi-empirical methods, physics-based models, machine learning models, and hybrid approaches, can be employed across these chemistries. However, the methods require calibration, fine-tuning, and validation for each specific battery type. It is believed that SoS holds significant potential for maritime applications, provided it incorporates a relevant set of safety sub-functions with properly defined thresholds and warning criteria. Its integration into real-time monitoring systems appears feasible, given continuous measurement of relevant inputs. However, further research is recommended on how to best account for interdependencies between the various safety sub-function and correlations in the input data as well as how to account for the effect of degradation on SoS. Additionally, it seems reasonable to investigate whether some kind of memory could be incorporated in order to account for the experience of previous abusive conditions.

Introduction and background

Secondary batteries degrade over time due to many factors, see e.g. [1,2], which means that their capacity to deliver energy and power deteriorates over time. State of Health (SoH) is a measure that can be related to either capacity fade, internal resistance increase or both. It is important to accurately monitor the state of health of maritime (SoH) battery systems used for propulsion and maneuvering. Together with state of charge (SoC), this enables the crew onboard to monitor the available energy in order to ensure that the vessel has sufficient energy and power for the intended operation. Failure to ensure this might leave the vessel unable to complete its intended voyage and lead to serious accidents such as collision and grounding. Because of this, maritime battery systems are required to have a battery management system (BMS) that, among other things, continuously estimates SoH. However, it is acknowledged that SoH estimates from the BMS may not be accurate or reliable. Therefore, for ships relying on battery power for propulsion, classification societies require the SoH estimates from the BMS to be verified by independent tests [3,4]. It is not specifically prescribed how this independent verification should be carried out, but this can normally be done by an annual capacity test where the vessel is taken temporarily out of service to perform a physical test. The batteries are then slowly charged to its full capacity and then slowly discharged under nominal, controlled conditions. However, this procedure is expensive and time-consuming and the results are not necessarily accurate.

Data-driven estimation of SoH based on sensor data is an alternative means of performing the independent verification of SoH estimated by the BMS. The overall idea is that sensor data of temperature, voltage and current can be collected continuously from the ship and fed into appropriate algorithms to predict SoH. These models obviously need to be different and independent from the algorithms running on the BMS, and the need to be reliable and accurate enough to yield equally robust estimates of SoH as one would have achieved by a physical test.

Data-driven approaches to SoH estimation for maritime battery systems were thoroughly investigated in a recent research project.

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A literature review identified a number of different data-driven approaches and concluded that accurate and reliable estimation of SoH remain a challenge [5]. Furthermore, several approaches were tried out and evaluated on various battery datasets [6,7]. One method was found sufficiently robust to replace the need for an annual capacity test, given that some operational conditions are met (e.g. experiencing deep enough cycles) and conditional on sufficient characterization data being available. A brief summary of these approaches and main results will be given below. However, only one particular type of lithium-ion NMC (Nickel Manganese Cobalt) cells were investigated. It is known that different battery types degrade differently, and the objective of this paper is to discuss data-driven SoH estimation for alternative battery chemistries. This includes lithium-ion batteries with different composition as well as non-lithium types of batteries, and the main focus will be on LFP-type (Lithium Iron Phosphate) lithium-ion and Na-ion batteries. This work focuses on LFP lithium-ion and sodium-ion batteries mainly due to their promising applications in the maritime domain (e.g., ferries, tugboats, and coastal vessels). LFP batteries are widely recognized for their exceptional thermal stability, safety, and long cycle life, making them a mature and reliable choice for energy storage and zero-emission solutions in the maritime applications. Sodium-ion batteries, on the other hand, offer sustainability advantages and resource abundance, addressing concerns about the limited availability and cost of lithium. It is worth mentioning that other types of batteries such as solid-state and zinc-ion batteries also show great promise but their current development is not mature enough to be applied for near-term maritime applications due to high production costs and limited scalability, and therefore is not included in the scope of this work. Moreover, existing work [8,9] also indicated that existing methods such as data-driven methods were mainly developed to estimate the SOC of solid state batteries, which could not be sufficient and accurate for solid state battery estimation. One reason to argue is that existing methods usually focus on the cell level rather than battery pack level, which can lead to inaccurate prognostications when aggregating.

It is noted that some studies have promoted transfer learning techniques for transferring machine learning models pre-trained on one type of batteries to batteries with other types of chemistries and charging protocols [10–13], see also the review in [14]. Hence, it is possible that models and algorithms developed for a specific type of NMC cells can be transferred to apply to other lithium-ion chemistries and non-lithium batteries. However, this would require a pre-trained base model, possibly trained on a vast amount of publicly available battery data, and additional representative training data for the target battery type. Note also that transfer learning has been suggested as a way to account for differences in operating conditions and for example to apply models trained on laboratory data to batteries under normal operations [15–18]. An attention-based recurrent neural network model is proposed in [19] as a chemistries agnostic lifetime prediction model for batteries, that is reported to perform well for both lithium-ion and sodium ion batteries.

Most studies and applications of secondary batteries are from non-maritime applications, and there are not much literature on how ship-specific conditions influence the degradation of batteries. For example, energy storage systems onboard ships in operation might experience different vibration conditions and more movements compared to other applications [20], and previous studies have indicated that vibration may significantly accelerate battery aging [21,22]. Battery systems onboard ships may also be exposed to larger temperature variations than other applications [20], although battery rooms onboard ships should, in principle, be well temperature controlled. Salinity and humidity are other actors that might influence shipboard battery systems more than other applications. Such effects are not considered specifically in this paper, but it should be acknowledged that results from studies on state of health and state of safety estimation for batteries used for e.g. stationary storage and electrical vehicles may not be directly transferable to maritime applications without accounting for maritime-specific effects.

In addition to data-driven approaches to SoH estimation, this paper will also review and discuss definitions and methods for determining the state of safety (SoS) of batteries, with a focus on lithium-ion LFP types of batteries and sodium-ion batteries. The state of safety is a relatively new measure that has not previously been much used in maritime applications. It is noted that several other measures are used for describing the states of batteries [23]. For example, state of power (SoP) describes the maximum charge and discharge power that a battery can support [24], the state of energy (SoE) is a measure of the energetic reserve in a battery [25] and the state-of-latent-energy (SoLE) is an alternative to state of charge that does not require an estimate of state of health for normalization [26]. However, this paper focuses on state of health and state of safety.

Moreover, new computational methodologies have also been studied based upon artificial intelligence (AI) for the future materials and devices [27–29]. For instance, The study in [27] addressed the challenges of lithium–sulfur (Li–S) batteries, such as lithium polysulfide dissolution and low conversion efficiency, by designing homonuclear transition-metal dimers embedded in monolayer C2N. Through first-principles calculations. The work demonstrated that these materials can enhance the anchoring of lithium polysulfides, suppress shuttle effects, and improve catalytic conversion efficiency, offering insights into dual-atom catalyst design for high-performance Li–S batteries. To improve the accuracy of short-term photovoltaic (PV) power prediction, the work [28] proposed a model integrating weather-based clustering, advanced data decomposition, and an enhanced Informer algorithm. By clustering data by weather conditions and employing precise data reconstruction methods, the model achieved improved accuracy and stability, validated through experiments with real-world PV plant data. The review [29] explored the development and potential of liquid–liquid triboelectric nanogenerator mechanisms and applications, which converts mechanical and tidal energy into electricity via liquid–liquid frictional interactions. It also discussed the fundamental principles, operational mechanisms, materials used, and applications in energy harvesting and medicine, while identifying challenges and opportunities for future advancements in this emerging nanogeneration field.

In summary, the core motivation behind this work stems from the growing adoption of alternative battery chemistries with a particular focus on LFP lithium-ion and sodium-ion batteries, in maritime applications, driven by the need for safer, more sustainable, and efficient energy storage solutions. Despite their potential, these chemistries lack comprehensive review for investigating and assessing critical performance metrics like state of health (SoH) and the emerging state of safety (SoS), which are essential for ensuring reliability and safety in demanding maritime environments. The paper aims to bridge this gap by reviewing the existing SoH estimation methodologies to the alternative chemistries and exploring the integration of SoS into real-time monitoring systems. Ultimately, the key contributions of this work include: (1) enhance the understanding of SoH and SoS for data-driven safety and health monitoring solutions for maritime battery systems and (2) outline the potential research directions for the relevant practitioners to work on (e.g., SoS).

Alternative battery chemistries

There are different types of rechargeable batteries, also referred to as secondary batteries. Previously, the most common secondary battery technology was lead–acid type of batteries. However, in recent years, lithium-ion batteries have been the preferred choice due to their increased efficiency, energy density and extended cycle life; see e.g. [30,31] for a thorough introduction to lithium-ion batteries, and see [32] for an overview of important aspects of maritime battery systems. More recently, other types of chemistries, utilizing non-lithium charge carriers, e.g. sodium-ion (Na-ion), zinc-ion (Zn-ion) or

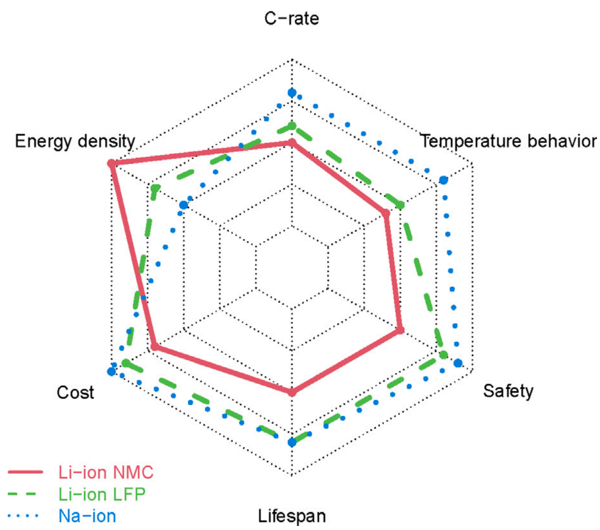

Fig. 1. Comparison of lithium-ion and sodium-ion battery cells within six dimensions.

magnesium-ion (Mg-ion) have emerged as alternatives to lithium-ion. In this subsection, a very brief review of different battery technologies and chemistries will be presented.

A comparison of selected characteristics of lithium-ion NMC, lithium-ion LFP and sodium-ion batteries is illustrated in the spider diagram in Fig. 1, reproduced based on [33]. It is obvious from this simple illustration that no technology is universally better, and that the preferred solution will be a trade-off between different considerations. An evaluation of commercial sodium-ion batteries presented in [34] suggested that the temperature behavior as a function of C-rate and cell impedance are comparable for sodium-ion and lithium-ion cells, and that the tested sodium-ion cells had a higher gravimetric energy density than tested LFP cells. A comparison of typical SoC-OCV curves for different chemistries is also presented in [33] (OCV = open circuit voltage; the terminal voltage without any load). One striking difference that is observed is that the OCV-curve is almost flat for lithium-ion LFP cells, a fact that is generally well acknowledged; see also [35–37], whereas the OCV-curve for Na-ion varies more than that for lithium-ion chemistries. This indicates a larger change in voltage during charging and discharging for sodium-ion cell, which may be an advantage if this is used for state estimation and if changes in OCV are used to model degradation. On the contrary, estimation algorithms relying on the OCV might not be suited for LFP cells due to the very flat OCV curve.

The main function of a secondary battery is to convert electrical energy into chemical energy for storage and then to release the energy by converting it back to electrical energy when needed. a secondary battery cell typically consist of a few main components: positive and negative electrodes (often referred to as cathode and anode), the electrolyte, a separator and current collectors. The cell's active materials reside in the electrodes where oxidation (loss of electrons) and reduction (gain of electrons) occur in order to liberate or bind positively charged ions and electrons. The liberated ions are allowed to diffuse between the electrodes through the electrolyte, and the electrons can be transported by the current collectors to generate a potential between the battery terminals and hence drive a current in an outer circuit. The separator should allow for transport of the charged ions but block electron transport to prevent internal short circuits. Most common batteries use liquid electrolyte solutions, but solid state batteries are battery cells with solid electrolyte. In lithium-ion battery cells, the electrical charge are carried by lithium ions (Li^+). In sodium-ion batteries, the charge carriers are sodium ions (Na^+), but the cell components and the electrochemical reaction mechanisms are similar to that of lithium-ion batteries. However, since Na^+ is larger and heavier than Li^+, a sodium-ion battery will typically have lower energy density compared to lithium-ion.

Different composition of the electrode and electrolyte materials yields different types of lithium-ion batteries. The most common types are NMC (Nickel Manganese Cobalt), LFP (Lithium Iron Phosphate), LTO (Lithium Titanate), NCA (Nickel Cobalt Aluminum), LMO (Lithium Manganese Oxide) and LCO (Lithium Cobalt Oxide). Even though these are all examples of lithium-ion batteries, they have different properties in terms of cost, gravimetric and volumetric energy density, cycle life, optimal voltage, current and temperature operating ranges and the need for scarce materials. They also respond differently to various stress factors with respect to aging, as presented in [38].

In addition to the large number of different lithium-ion types of cells, other types of batteries uses non-lithium material as charge carriers. Some examples of these are sodium-ion, zinc-ion and magnesium-ion, where sodium, zinc and magnesium, respectively, are used as charge carriers between the electrodes. These alternative chemistries will have different characteristics again, in terms of operational limits and degradation. It is out of scope of this paper to discuss this in detail, but it is noted that different types of lithium-ion and other types of batteries will degrade differently, and respond to variations in loading conditions and various stress factors differently. Also, safety aspects of batteries will be dependent on the battery chemistries, and different types of batteries will have different thermal behavior and respond differently to various abuse factors. In fact, it is noted in [39] that operating performance may differ significantly even for cells with the same underlying chemistries but from different manufacturer. A review and comparison of thermal behavior, degradation and safety between lithium-ion and sodium-ion batteries are presented in [40]. It is stated that very little is published on the degradation of sodium-ion batteries at cell level. At any rate, for different types of battery cells, one cannot tacitly assume that one may easily transfer data-driven models trained on one type of cells to another.

In this paper, an investigation into data-driven approaches to SoH estimation for the types of batteries will be presented. That is, the focus will be on estimating SoH for LFP lithium-ion and sodium-ion types of batteries. Since actual data from such types of batteries are scarce, this investigation is initially based on a review of the scientific literature on the topic.

Previous research on NMC lithium-ion batteries

Previous research has focused on establishing data-driven models for estimating SoH for particular types of NMC lithium-ion batteries. In this subsection, a brief summary of the methods and lessons learned from that study will be presented, although it is acknowledged that results cannot be directly transferred to alternative battery chemistries.

A comprehensive literature survey on data-driven methods for capacity estimation of lithium-ion batteries was performed [5]. This revealed a plethora of methods ranging from direct measurement techniques to simple semi-empirical and statistical models to more advanced machine learning techniques. Methods were further classified into cumulative and snapshot methods. An illustration of the categorization of different methods is shown in Fig. 2 (from [5]). Some of these approaches were further explored and tried out on various battery datasets.

The Battery AI tool [41], which is a cumulative model combining a semi-empirical model with machine learning was tested on maritime battery data, as reported in [42]. Even though the method could demonstrate reasonable results, for a subset of the data, this exercise revealed some limitations of cumulative methods in general. The analysis turned out to be extremely computationally expensive, due to the massive amount of data from very large maritime battery systems, consisting of a large number of cells and with frequent sampling. This issue could probably be improved by optimized data handling, but another more fundamental problem with cumulative methods was

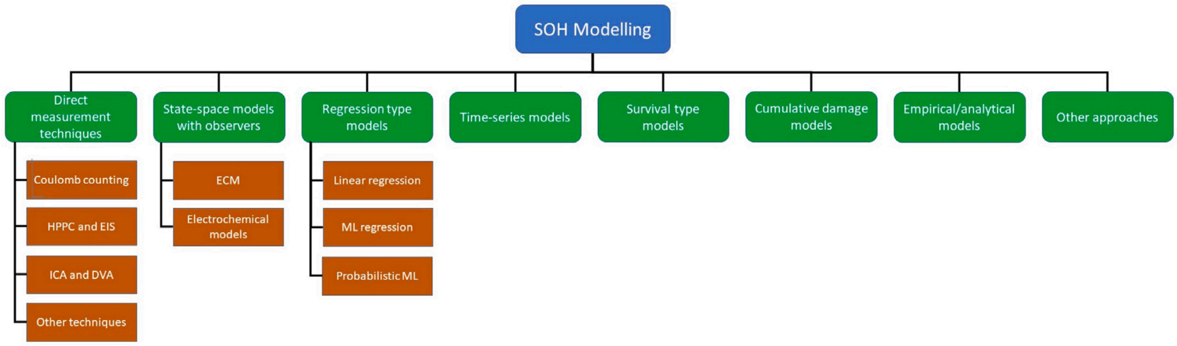

Fig. 2. Categorization of approaches (in total eight) to SoH modeling.
Source: From [5].

revealed: cumulative methods need the complete data history in order to estimate current capacity and state of health. However, the maritime battery datasets available for the study contained several long gaps in the data, and this might be difficult to avoid for maritime battery systems. This could possibly be improved by better and more reliable ship-to-shore connectivity, but it was deemed difficult to guarantee complete data history from ships in operation, and therefore cumulative approaches in general were not recommended.

An alternative and somewhat simpler cumulative approach, where the loading history was condensed in histograms or buckets, was explored in [43,44]. Various non-sequential and sequential machine learning models were fitted to such histogram data, and results were reasonable, but it would still suffer from the need for complete data history in order to estimate SoH and was therefore not recommended for maritime battery systems. Also, the multivariable fractional polynomial regression models presented in [45] are cumulative in the sense that they estimate the change in capacity rather than actual capacity from the selected features.

Realizing that cumulative models would be associated with practical challenges, different snapshot methods were explored. Various snapshot features were extracted from charge and discharge curves and used to establish simple statistical regression models in [46]. The methods were trained on data from laboratory cycling tests. Although results were promising, they were not accurate enough, and the main reason might be lack of sufficient relevant training data. However, even with an extended dataset this remains a challenge for such methods, as presented in [47]. It was concluded that for the snapshot methods to perform well, the loading conditions that the cells to be predicted experience need to have been experienced in the training data, and this might be difficult to guarantee. Typically, models that needs to be trained require laboratory cycling data, which are expensive and time consuming to generate, and it will be difficult to design experiments that ensure that every imaginable loading conditions will be well represented. More complicated machine learning techniques were used to establish snapshot methods in [48], that was found to perform reasonably on laboratory cycling data. However, the models were not applied to actual operational data and the challenge of obtaining representative training data would remain.

The voltage deviation method is another snapshot approach that was explored [49]. This estimates capacity by exploiting a presumably strong correlation between increase in internal resistance and decrease in battery capacity. This correlation might be strong for some battery types, but not necessarily for others. At any rate, this model also needs training and it was found challenging to obtain sufficient relevant training data for this method to yield sufficiently good results.

Obtaining sufficient relevant training data from laboratory tests is expensive and time-consuming, and the resulting data might not be representative of actual operational conditions. Hence, it was investigated whether operational data from existing battery installations could be used for training. However, one problem with this approach is that operational data contains very few labels or actual measurements of battery capacity. Typically, labeled data are only available once per year when results from annual capacity tests are performed. Hence, various semi-supervised approaches to estimate SoH were explored, where operational data from existing battery installations are used for training after labeling [50]. However, also this approach suffers from some fundamental challenges. For example, it is difficult for such models to predict a larger extent of degradation than what has been experienced in the training data, which means that ideally operational data from several vessels with batteries that have reached end of life would be required. Moreover, operational conditions not experienced in the training data would be difficult to predict. Hence, also this approach has limitations.

Due to several challenges with purely data-driven approaches to SoH estimation, it was concluded in previous research that this might not be enough without incorporating some physics-based elements in the modeling. Hence, simple linear models based on Coulomb counting were explored in [51,52], inspired by the works presented in [53]. Again, results were promising but not accurate enough, partly due to large variability in loading conditions, and the fact that SoH and battery capacity is not a fixed quantity; it will depend on charging and discharging conditions. Results could be improved somewhat by data filtering and pre-processing and by applying an ensemble of such linear models for different subsets of the data, as described in [54]. However, results were still not entirely satisfying. One of the reasons for this is probably that the Coulomb method depends heavily on the state of charge (SoC), which is a derived quantity and not directly measured. Although it is assumed that estimating SoC is less challenging than SoH, it is still associated with notable uncertainties, and therefore it is problematic to establish data-driven methods based on this as a feature.

In order to minimize the dependency of SoC, an extension of the simple linear model based on the open circuit voltage (OCV) was developed. With this approach, an equivalent circuit model was introduced to obtain the open circuit voltage from current, temperature and voltage measurements. Then, the open circuit voltage is mapped to SoC using a known cell-specific OCV-SOC curve. This can e.g. be established by specific characterization tests. Results from this approach can be found in [55], which unfortunately still display some lack in precision.

One of the main reasons that was identified for the lack of precision might be the failure to accurately account for variations in operational conditions. For example, temperature effects and hysteresis were not well described by the model. Hence, a further extension of this approach was proposed where an improved equivalent circuit model that includes a hysteresis model and a thermal model was introduced (see e.g. [7,55]). Additionally, several lookup tables based on extensive characterization tests were used in order to account for variations in battery states from variations in operational conditions such as variations in temperature and current. Preliminary validation of this method, on both laboratory and field data, indicates that is can provide reliable and accurate estimation of SoH from operational data, conditional on the battery system having experienced sufficiently deep cycles. Further validation of this method is ongoing, and the required deepness of the operational cycles will presumably be battery-type specific. Notwithstanding, with this approach, data-driven verification of SoH as an alternative to annual capacity tests has been made possible.

The methods that were previously explored as summarized above were all tried out for NMC-type of lithium-ion battery systems. Since it is well known that different battery types and chemistries degrade differently, and respond differently to various stress factors, it is not obvious that the methods can be easily transferred to alternative cell types. For example, LFP types of lithium-ion batteries are known to have very different characteristics of the charge and discharge curves compared to NMC types of batteries: LFP cells generally have flat charge/discharge curves compared to NMC. While this might gives some advantages such as the ability to deliver constant power over an extended period of time, it also makes it more challenging to determine SoC based on voltage level. In particular, this means that methods that are found to work well for NMC lithium-ion battery cells that relies on SoC (directly, or indirectly via an SOC-OCV curve), may not work as well for LFP cells. This may rise the need for completely different types of data-driven models. Also, the degradation of Na-ion cells might be different to that of Li-ion, and it is indicated in [56], that Na-ion has higher cyclability and might not experience a knee-point in the degradation trend.

In this paper, these differences will be discussed, and a literature survey will be presented on data-driven SoH methods for specific chemistries will be presented, with a focus on LFP Li-ion and Na-ion batteries.

Data requirements

Relevant data is obviously needed for data-driven methods. If the aim is to establish data-driven methods and algorithms to predict battery degradation and estimate state of health, then obviously data measuring battery degradation over time is needed. The specific data requirements will depend on the actual methods and algorithms one wants to develop, but obviously data from cells that have experienced notable degradation are needed to train the algorithms — ideally covering the cell's entire lifetime, from beginning of life until past end of life. If data-driven methods are to detect and predict battery failures and associated risks, then obviously battery failure data are needed.

Moreover, it is generally known that different battery cells degrade differently, and might respond different to various stress factors. Hence, it is important to have training data from identical, or at least similar, battery cells to the cells one wants to apply the data-driven methods to. Generic cycling data may not be representative for cells even with similar chemistries and compositions. Hence, not only is there a need for long time series, but the data need to be specifically for the type of cells in question. Even with techniques such as transfer learning which could exploit more generic battery data, there would need to be some amount of cell-specific data.

It should also be acknowledged that various types of data-driven approaches have different data requirements [7]. For example, if cumulative methods are used [42], then data from the full loading history might be needed; possibly compressed in the form of histograms [43]. However, if snapshot methods are to be exploited, then full histories might not be needed, and high-resolution data of relevant features at certain intervals might suffice [47]. Moreover, purely data-driven models typically extrapolate poorly, so data might need to be combined with physics-based models, or data need to be covering all realistic combinations of operating conditions. Hence, the data requirements need to be assessed carefully considering the types of models that are to be used.

One way of obtaining relevant training data is to perform laboratory cycling tests. However, this is expensive and time consuming and it cannot be expected that sufficient lab data will be available. An alternative is to exploit operational data, if they exist. However, this would require that the battery cells and systems in question have already been in use for a considerable time so that significant degradation and drop in SoH has occurred, and at the same time that reliable SoH measurements are included in the data. This will seldom be the case, and particularly not for novel systems with alternative battery chemistries. It is also challenging to combine data from different battery systems that have experienced very different usage patterns [57].

For the purpose of testing out different data-driven methods within a research environment, publicly available datasets represent a possible source of training data. Hence, in the following, a review of some publicly available battery dataset will be given, with a focus on cycling data from LFP and Na-ion cells.

Publicly available battery data

A recent overview of available datasets for lithium-ion batteries is presented in [58]. More than 30 datasets are discussed, covering cycle and calendar aging datasets, and both testing data from laboratory experiments and synthetic data obtained from augmenting existing data or from generating artificial data. The datasets presented in [58] are for different types of lithium-ion batteries, and NMC/LCO/NCA are the most common chemistries. However, a few datasets for LFP types of batteries are presented, e.g. from Toyota Research Institute (https://data.matr.io/[1]), from Sandia national laboratories (https://www.batteryarchive.org/snl_study.html), from Mendeley data (https://data.mendeley.com/datasets/r4n22f4jfk/1 [59]; https://data.mendeley.com/datasets/bs2j56pn7y/1 [60,61]; https://data.mendeley.com/datasets/6s6ph9n8zg/1 [61,62]) and from the University of Science and technology of China [63]. Another review of lithium-ion datasets is presented in [64]. A methodology for generating big-data training datasets for intercalation batteries is proposed in [61]. It combines modeling with experiments to provide a generic tool for creating synthetic voltage curves. Proof-of-concept datasets were generated for both diagnostics and prognostics purposes for a commercial LFP battery.

Some of the publicly available data repositories presented in [58] for cycle data are the Battery archive (https://www.batteryarchive.org/), the energy storage safety data repository (https://www.sandia.gov/energystoragesafety/rd-data-repository/) and the National Renewable Energy Laboratory (NREL) datasets (https://www.nrel.gov/research/data-tools.html); see [58] for the full list. Battery data from NASA have been thoroughly analyzed by several authors, see [65], and data from accelerated life testing described in [66] can also be found here. Some additional battery data repositories not mentioned in [58] include the automated battery database (https://aging.battronics.com/database/0), the data used in [67] (https://data.matr.io/1/projects/5c48dd2bc625d700019f3204) and the data used in [68] (https://data.matr.io/1/projects/5d80e633f405260001c0b60a). The list of publicly available battery datasets are presumably growing and it is advised to perform a new search for data when the need for data arises.

More focused on thermal runaway behaviors of lithium-ion batteries, the battery failure databank (https://www.nrel.gov/transportation/battery-failure.html) provides datasets from several abuse tests of batteries [69].

As far as the authors of this paper are aware, there are currently no publicly available datasets for sodium-ion batteries, although this might be expected to become available as the technology mature and become more widely used.

[1] This and the other URLs presented in the following were accessed on 9/12-2024.

Commercially available models

Some battery degradation models are commercially available for various types of battery cells. It is not the aim of this paper to provide a comprehensive overview and assessment of these, and the models have not been tested, but it should be mentioned that such products exist. One example is the suite of TWAICE battery aging models for lithium-ion batteries [70], which are based on an equivalent circuit model and a semi-empirical aging model. Three variants of the models are the generic base model, the customized base model and the premium model with increasing levels of being cell-specific. A simulation model for sodium-ion batteries is also available from TWAICE [33], based on the same approach as the model for lithium-ion cells.

Battery AI is a battery model provided by DNV that is based on a combination of a semi-empirical model and a machine-learning model for predicting battery degradation [41]. It is essentially a cumulative model that adds the degradation contribution from repeated cycles under varying conditions. It has been tested on maritime battery data in [42]. It has been trained on various lithium-ion types of batteries (including both NMC and LFP), but not yet on sodium-ion batteries, as far as the author of this paper is aware.

MathWorks offers battery simulation models, including estimation of state of charge and state of health, with Simulink and Simscape [71]. It is noted that there are no universal solution for defining SoH, and that there are therefore no general-purpose, off-the-shelf solutions for estimating SoH, but that Simscape Battery provides an SoH estimator block using a Kalman filter [72].

PyBaMM is a battery model for Python that includes simulation models for battery performance and aging [73]. Its library includes a wide range of physics-based models including various electrochemical lithium-ion models and equivalent circuit models. An extension of this for simulating lithium-ion battery packs is available in the liionpack package [74]. PyBaMM is also used to build the model in [75] that couples several degradation mechanisms. ParaSweeper is a Python script based on PyBaMM for high-throughput computing of complex physics based degradation models for lithium-ion batteries [76].

The Battery and Electrochemistries Simulation Tool (BEST) from Fraunhofer is a software for modeling lithium-ion batteries based on a physics-based, three-dimensional multiscale model [77]. Other battery models and tools include GT-Autolion and GT-Suite from gamma technologies [78], Batemo cell models [79], CellSage [80] and Battery Cell Designer in Simcenter from Siemens [81]. This list is not exhaustive and it is merely mentioned that there exists a plethora of commercial software tools for modeling and simulating secondary batteries. However, capacity degradation and aging of rechargeable batteries remains challenging, and countless of scientific papers have been published addressing this, even though commercial software is available.

Data-driven state of health monitoring

Previous research explored a number of different data-driven approaches for SoH monitoring of lithium-ion batteries, i.e. for particular types of NMC li-ion cells. In addition to this, there are a lot of literature on the degradation of batteries, but most focuses on lithium-ion batteries; very little is published on the degradation phenomenon of sodium-ion batteries. A comprehensive review of degradation modeling is presented in [5], and numerous other reviews exist, but mostly focusing on lithium-ion. A comparison of degradation for lithium- and sodium-ion batteries is presented in [40]. Some of the findings reported here are that SEI (solid electrolyte interphase) instability might be more pronounced for Sodium-ion than lithium-ion and that for sodium-ion batteries, cycling at low C-rates might cause increased degradation, but that sodium-ion batteries are less prone to plating and dendrite formation during fast charging [82]. Cycling tests reported in [83] confirm that there are no strong effects of high charge currents on capacity fade of sodium-ion batteries. However, it is noted in [40] that very little literature is available on cell-level comparison of lithium- and sodium-ion degradation and aging, and that more investigations are necessary to understand the aging mechanisms in Na-ion batteries under normal operating conditions. Moreover, as highlighted in [84], most research focus on cell-level diagnostics and ignores the additional uncertainties associated with module, pack and system level safety and performance, something that is a naive oversimplification. Hence, it is recommended to focus on diagnostics for whole battery systems, see also [85]. Notwithstanding, in the following sub-sections a review of literature on data-driven SoH estimation particularly for LFP Li-ion and Na-ion types of battery cells will be presented.

SoH estimation for Li-ion LFP

LFP is an established technology which has been used in many applications, including stationary storage and electric vehicles [39]. Results of some cycling tests with both LFP and NMC types of cells are presented in [39], indicating that the capacity degradation of different LFP cells might be quite variable even for the same cycling conditions. However, results were less variable when cycled with reduced C-rates. With regards to the effect of temperature on capacity degradation, results indicate that LFP cells may be less influenced by temperature compared to NMC; NMC cells experienced very little degradation when cycled in room temperature but notably more degradation at increased temperature whereas LFP cells exhibited notable degradation at both temperatures, albeit to a larger extent for the increased temperature. Moreover, LFP cells seems to be more affected by the average SoC during cycling compared to NMC. However, it should be emphasized that these are observations from a limited number of tests only, and one should be careful with basing general conclusions based on this. It is stated in [39] that even though it may be possible to draw some general conclusions regarding degradation for a given chemistries, this will not tell the full story for a specific cell model, and laboratory testing for specific cells may be important.

The relatively flat OCV curve of LFP cells, corresponding to a wide constant potential plateau during charge and discharge, makes it challenging to estimate SoC from OCV for LFP batteries. Hence, methods relying on SoC for estimating SoH might be less accurate for LFP batteries compared to NMC types of batteries. Moreover, the analyses performed in [7] illustrated to difficulty in relying on SoC even for NMC battery types. Hence, it is believed that OCV-based approaches will not be precise and accurate enough for LFP cells. On the other hand, the method proposed in [86] utilizes the length of the voltage plateaus in order to estimate capacity, and claims that this contains sufficient information, without the need to fully discharge the battery. Hysteresis effects can also be notable for LFP batteries and are affecting the open circuit voltage.

Empirical and semi-empirical models are often used for battery degradation. However, a serious challenge with such approaches is that battery degradation is path dependent, i.e., cell degradation is dependent on how they have been used. A semi-empirical aging model based on ampere-hour throughput with corrections for variations in temperature, state of charge and C-rates, and modified to account for dynamically changing conditions, is applied to both NMC and LFP lithium-ion cells in [87]. Results demonstrate that such a model yields reasonable results for both NMC and LFP chemistries. However, it is based on infinitesimal changes over time, and hence suffers the problems of any cumulative damage models. It requires complete operating histories to obtain estimates of the capacity. Numerous other semi-empirical models for aging of LFP cells are proposed in the literature, see e.g. [88–91]. A critical review of semi-empirical approaches to model capacity fade is presented in [92] which points out that semi-empirical models are faced with serious limitations. The comparison presented in [93] reports that physics based electrochemical models tend to outperform empirical and semi-empirical models for LFP cells, but at a computational cost.

Equivalent circuit models (ECM) is often used for state estimation of batteries. A first-order resistance-capacitance (RC) equivalent circuit model is proposed to co-estimate SoC and SoH in [94], and is reported to perform better than conventional OCV-based methods. The effect of SoH, SoC and temperature on the model parameters for LFP cells was investigated in [95], and an empirical model was developed to account for this. A third-order RC model integrated with a hysteresis model is proposed for LFP batteries in [96], with a focus of estimating SoC. This highlights the need for accounting for the hysteresis effect of LFP batteries. Electrochemical models, also referred to as physics-based models (PBM) ranging from 1D to 2D and 3D models, are alternatives to equivalent circuit models that can be used to model aging of batteries. They may describe capacity degradation due to loss of lithium inventory (LLI) and loss of active material (LAM). A simplified electrochemical and thermal model is proposed in [97] for power and capacity fade simulations, see also [98,99] for other coupled electrochemical–thermal models. A comparison of electrochemical–thermal models for describing lithium-ion batteries is presented in [100], suggesting that 3D models perform best. However, a drawback with electrochemical models is that they are computationally costly, and increasingly so for more complex models. A comparison of equivalent circuit models and physics based models for LFP batteries is presented in [101]; ECMs are typically easier to calibrate, less computationally demanding and accurate within the calibration range for some current profiles, whereas PBMs can maintain its accuracy beyond the calibration limits but are more computationally demanding and requires more laborious calibration to determine a large number of parameters. It hence is suggested in [101] that ECM and PBM are complimentary tools for modeling and simulating realistic battery operations.

Since battery degradation is path-dependent, adaptive models, e.g. based on Kalman filters, particle filters, neural networks, fuzzy logic or other data-driven approaches has been proposed. In principle, adaptive models may be accurate, but they do require extensive training data to estimate the correlation between model parameters and cell degradation, and they might be very computationally intensive. An alternative approach is proposed in [102], which identifies some features of interest from incremental capacity analysis relating the cell voltage-response to degradation and generate a multi-dimensional lookup table based on a large number of simulated degradation paths. This approach, based on lookup tables has a low computational cost in operation, but requires prior experimental testing and simulation in order to establish the tables. These first steps are computationally costly. Results are reported for NMC and LFP lithium-ion cells, but is said to be chemistries agnostic. However, it is noted that diagnosis can be difficult for chemistries with large voltage plateaus, which is typical for LFP batteries. An SoH algorithm based on cell inconsistencies and Lorenz plots, where the average Lorenz radius of a module is used to estimate module SoH is presented in [103] and applied to LFP-type batteries. A linear relationship between SoH and the average Lorenz radius was found, and the algorithm shows good goodness-of-fit for LFP cells connected in series and parallel.

A number of data-driven approaches relies on machine learning techniques for SoH estimation. These would typically extract features from the data and model the relationship between these features and the battery capacity or state of health. A large range of different methods are reported in the literature, using different features and information. A data-driven approach to extract aging features from battery data is presented in [104], with the aim of improving the performance of machine learning algorithms. The feature extraction is based on a combination of algorithms and when applied to LFP cells, an improvement in accuracy of SoH estimation up to 71.1% is reported. It is also reported to work well on features extracted from partial voltage ranges. A different approach to feature extraction for estimation of SoH is proposed in [105], where features are selected theoretically based on their relationship with SoH. Features extracting from charging curves and temperatures are selected and validated based on Grey Relational Analysis theory and used to generate training data for data-driven methods. A generative deep learning model is used in [106] to extract aging-related features from data to generate synthetic OCV curves. A methodology to generate large amounts of synthetic battery data for training data-driven models is proposed in [61]. In addition, the Kolmogorov–Arnold Networks (KAN) method represents a state-of-the-art approach for battery State of Health (SoH) estimation, leveraging its ability to model complex, high-dimensional, and nonlinear relationships with mathematical precision [107]. Rooted in the Kolmogorov–Arnold representation theorem, KAN decomposes multivariate functions into sums of univariate functions, facilitating accurate approximation of battery degradation behaviors. Recent advancements in KAN frameworks have improved their scalability and computational efficiency, enabling real-time SoH predictions under dynamic operational conditions. For instance, a SOH estimation method combining Kolmogorov–Arnold Networks (KAN) and Long Short-Term Memory (LSTM) networks was proposed in [107] by extracting parameters such as voltage and temperature and the results demonstrated that higher SoH estimation can be achieved by employing KAN methods.

A large volume of recent literature on the use of machine learning for battery capacity degradation prediction is available. For example, Convolutional neural networks are proposed in [108], where battery data are represented as images, to model battery degradation. The method is reported to perform well on tested on data from real LFP cells. A feedforward neural network is used in [109] to estimate SoH of LFP batteries cycled at various power levels. The hybrid deep learning framework presented in [110] combines both convolutional neural networks and recurrent neural networks to estimate SoH. Neural ordinary differential equations based machine learning models are used to predict state of health in [111] and compared to some standard recurrent neural networks. A deep neural network is suggested in [112], which was found to outperform an alternative model based on a feedforward neural network. Deep learning is also employed in [113] to estimate state of health for different types of batteries. Other types of machine learning models that have been used for SoH estimation of LFP-type of lithium-ion batteries include Gaussian processes regression [114,115], support vector machines [116–118] and regression trees [119], a well as multi-model ensembles [120,121]. The integration of physics-based models and machine learning have also been suggested as an emerging technology for battery safety and management, see e.g. the review in [122]. It is out of scope of this paper to review these in detail, but suffice to say that a large number of machine learning models have been suggested for data-driven capacity estimation of batteries, including LFP.

SoH estimation for Na-ion

Sodium-ion batteries are, similarly to lithium-ion batteries, relying on intercalation and deintercalation of the charge-carrying ions at the electrodes. Hence, in many ways, the fundamental principles are similar for Na- and Li-ion cells, and many approaches to modeling the cells and their degradation mechanisms could be similar. Factors contributing to aging in sodium-ion batteries are discussed in [123], and although the actual degradation rates will be different for different battery types, many of the degradation mechanisms are similar for sodium-ion and lithium-ion. It can therefore be assumed that similar approaches to SoH modeling for li-ion cells can also be applied to Na-ion cells, after appropriate adaptation and fine tuning, even though the underlying degradation mechanisms might be very different. That is, it can be expected that empirical/semi-empirical models, ECM and electrochemical-based models and machine-learning based data driven models for capacity estimation should work equally well for Na-ion cells as for Li-ion cells, but obviously adapted to the particular cell. It is also stated in [83] that the transferability of diagnostic algorithms from lithium-ion to sodium-ion batteries are acceptable and that characterization methods for sodium-ion cells can benefit from established methods for lithium-ion cells. Data-driven methods depend on data, for training, calibration and fine-tuning, and there might be less available data on sodium-ion batteries for training and calibrating models. One possible way to handle this could be to exploit transfer learning and use models trained with large amounts of lithium-ion battery data and only relatively small amounts of sodium-ion and cell-specific data, as suggested in e.g. [10,14,19]. However, in the following, a brief review of methods reported in the literature specifically for sodium-ion batteries will be presented. Since sodium-ion batteries are a relatively less mature technology than lithium-ion, there is less experience with the former and the amount of literature is considerably less.

A review of battery degradation for lithium- and sodium-ion batteries is presented in [40]. It is stated that although many degradation studies on lithium-ion batteries are reported, there is very little literature on the degradation of Sodium-ion batteries. Hence, there are insufficient information on cell-level to compare the performance and aging of lithium-ion and sodium-ion batteries. One of the differences mentioned is that the solid electrolyte interface (SEI) might be less stable for sodium-ion batteries compared to Li-ion cells, which is a critical factor for battery degradation since it consumes cyclable Na^+ ions. One of the first studies on Na-ion cell degradation is presented in [82], suggesting that temperature is a key factor influencing cell degradation. However, they also found that low C-rates caused increased degradation during cycling, and that continuous intercalation — deintercalation of Na-ion cells induces mechanical stresses that might cause detachment of active material from the current collectors. It is also suggested that sodium-ion batteries may be less prone to plating and dendrite formation during fast charging, and that some types of Na-ion batteries could be good alternatives for applications requiring fast-charging. It is reported in [56] that the cyclability of sodium-ion batteries are comparable or better than lithium-ion batteries, and that the cyclability is good over a wide range of ambient conditions; specific sodium-ion cells tested under specific conditions were reported to last twice as many cycles compared to LFP cells cycled at similar conditions. Moreover, it was suggested that Na-ion batteries might not experience a knee-point in the degradation curves.

A study presented in [124] found some differences between sodium- and lithium-ion batteries, e.g. that hysteresis effects are only occurring at low SoC for Na-ion, that Na-ion does not exhibit long voltage plateaus in the OCV-SoC curve and that there is no obvious capacity regeneration during aging for Na-ion. This is advantageous for battery state estimation, including estimation of SoC and SoH. In particular, the near-linear relationship between OCV and SoC for sodium-ion batteries should in principle make SoC estimation easier, and requiring much less data and less complicated models [125].

Equivalent circuit models (ECM) can be used for sodium-ion batteries, as for lithium-ion. ECMs for sodium-ion batteries with different numbers of RC-elements are investigated in [126]. It is concluded that a third-order model is sufficient to describe the dynamic non-linear voltage response of the sodium-ion cell, and that this works better than lower-order alternatives. However, the ECM was not used to estimate SoH. Different SoC and SoH estimation methods for sodium-ion batteries are compared in [124], where the battery is modeled by a third-order ECM. Different algorithms are compared for estimation of SOC, i.e., extended Kalman filter, unscented Kalman filter and particle filter, and various regression models, i.e. multiple linear regression, ridge regression and support vector regression, are tested to estimate SoH based on three extracted health indicators. The extracted health indicators are constant current charging time, constant voltage charging time and constant current discharging time. These indicators may be extracted from testing data, where particular CCCV (constant current constant voltage) charging schemes are employed, but it is noted that they are not likely to be found in real operating data from batteries during variable loading. Moreover, the effect of e.g. temperature is not accounted for. Notwithstanding, it is reported that the methods achieve good results for sodium-ion cells, and that the best data-driven method is ridge regression.

Physics-based, electrochemical models can obviously also be used to model sodium-ion batteries. Physics-based models for complete Na-ion batteries are presented in [127,128]. A reduced order electrochemical model for sodium-ion batteries is presented in [129] as a quicker and less computationally costly alternative to pseudo-two dimensional electrochemical models. However, the models are proposed as useful design-tools and are not used to model battery degradation.

Machine learning have also been used for capacity prediction of sodium-ion batteries. The performance analysis presented in [130] applied machine learning to a large dataset constructed from the literature, and reported that random forest models are reasonably good in obtaining crude predictions for discharge capacity. Several other machine learning techniques were explored, but the scope of this study is somewhat different from on-line monitoring of capacity and state of health. Rather, average discharge capacity and cycle life performance are modeled as a function of various battery and electrode characteristics across different types of cells. Thus, this application of machine learning on sodium-ion batteries is directed more towards the design of the batteries than to real-time monitoring. A simple hierarchical learning model based on a pair of multi-layer perceptron neural networks is proposed in [125] for modeling sodium-ion batteries and estimate SoC, without the need for equivalent circuit or electrochemical models. Input data are current, voltage and temperature, and a low-complexity model is found sufficient due to the absence of voltage plateaus such as the ones observed for a typical LFP ell. A fusion of deep learning models is used in [131] to improve the modeling of state of charge for Na-ion batteries, and is reported to perform well under various operation conditions. Although these models are used for state of charge rather than state of health, it is acknowledged that SoC and SoH are dependent, and accurate SoC is needed for many approaches to SoH estimation.

The chemistries agnostic battery degradation modeling proposed in [19] is an attention-based recurrent algorithm for neural analysis with LSTM (Long short-term memory) architecture initially trained on a very large dataset from different sources, including proprietary and public data sources. It is reported that the model generalizes well over different chemistries, battery designs and cycling conditions, and it is demonstrated to perform also for sodium-ion batteries. It is designed to be applied to raw data without the need for prior feature engineering and includes an encoder–decoder framework based on LSTM. Two variants of this model is adapted and fine-tuned for Na-ion cells in order to predict battery degradation and monitor SoH. A recurrent neural network architecture is utilized to predict state of health in [10] lithium-ion and sodium-ion batteries. Then renormalization and normalization parameter training were performed in order to use a model trained on lithium-ion batteries for SoH estimation of sodium-ion batteries. Note that this involves pre-processing of the data before it is sent to the machine learning models and not an update of the model parameters. Results indicate that such a model can capture the main degradation tendencies, but with a notable bias. It is also noted that a reference model trained on Na-ion data performs better on the Li-ion data than a reference model trained on Li-ion applied to Na-ion data. However, this was thought to be due to differences in charging protocols rather than differences in chemistries.

The literature on data-driven state of health monitoring specifically for sodium-ion batteries is relatively scarce and the technology is still immature. Nevertheless, the literature survey suggests that there should be no fundamental differences between lithium-ion and sodium-ion batteries with regards to what type of approaches can be used to model capacity degradation and monitor state of health. It is believed that both empirical/semi-empirical models, ECM and electrochemical based models as well as machine learning approaches could be useful. Obviously, actual degradation will be cell-specific, so the models needs to be trained, fine-tuned and calibrated for the specific battery type it should monitor, but this is not different from the case with lithium-ion batteries. Moreover, some studies that have investigated transfer learning methods suggest that it can be possible to transfer data-driven models trained on one type of batteries (e.g. a type of lithium-ion) to other types (e.g. sodium-ion).

Data-driven state of safety monitoring

The concept of a measure for the state of safety (SoS) is relatively novel and not well established. In particular, it is not widely used in the maritime industries. However, as the safety of maritime battery systems is of uttermost importance, in particular from a classification society point of view, such a measure is deemed highly relevant. This section provides an introduction to the concept of state of safety (SoS) for batteries and battery systems, and a preliminary assessment of the feasibility of introducing data-driven state of safety for all-electric maritime vessels. It includes a literature survey on scientific literature on battery state of safety, which is currently scarce. It is noted that the discussions in this section is chemistries-agnostic, although applications reported in the literature are mostly focusing on lithium-ion batteries. An overview of safety aspects for sodium-ion batteries is given in [123], including how aging is reducing resilience to abuse conditions and hence affects safety.

There are currently no international regulations specifically for the safety of onboard battery systems, and the safety of such systems are left to classification requirements and industry standards. However, the European Maritime Safety Agency (EMSA), recently published a non-mandatory guidance document with essential functional safety requirements for lithium-ion energy storage systems on board ships [132]. Fire safety and the safety of electrical installations are main focus areas of these guidelines, which are developed based on IMO guidelines for developing goal-based standards. Several recommendations relating to the design and arrangements of the energy storage system is made, and means to detect, contain and suppress/extinguish fires both inside and outside of the battery room are suggested. Some explicit goal is that thermal runaway is detected as early as possible and contained at the lowest level avoiding propagation to other modules and that gassing and fire are detected early. This should be achieved by installing off-gas detectors and temperature sensors, with interfaces to the BMS for shutdown and isolation of unsafe battery modules. It is further stated that abnormal conditions related to high or low ambient temperatures, detection of gas, heat and smoke and failure of the detection systems should initiate an alarm. Other goals and requirements are related to e.g. ventilation, electrical arrangements and temperature control. It is further required that alerts and indicators should be raise alarms at a manned control position in case of e.g. earth fault detection, high or low battery voltage, high or low temperature, high charge current, and failure of the BMS or any sensors. However, there is no mention of a metric such as the state of safety (although it is mentioned as part of the definition of state of health).

A concept intimately related to battery safety is that of safe operating limits and the fact that safety is impaired by operating the battery outside various thresholds in voltage, current, temperature etc. Hence, control strategies for the safety of secondary batteries include making sure the battery is operated within the safety limits and thus to avoid abusive conditions [133]. A probabilistic approach to safer operating areas (SOA) is proposed in [134] based on abuses, failure mechanisms, electrochemical parameters and various health indicators. Large amounts of data and machine learning are used in [135] to define a safety envelope of mechanical loading conditions for lithium-ion batteries. Naturally, battery aging affects the safety limits of a battery, and this should be reflected in how the batteries are operated when the batteries age and approach end of life, see e.g. [136] for a recent review. Comparing lithium-ion and sodium-ion batteries, it is suggested in [123] that they have comparable safe operating windows with regards to temperature, but that sodium-ion has a wider safe operating window in terms of voltage and that they can tolerate lower voltages than lithium-ion batteries. However, this will obviously be dependent on the particular chemistries and other design factors.

Safety of batteries is typically associated with fire and thermal runaway [137]. Under certain conditions, external or internal, the energy stored in the batteries might be released at a rate faster than it can be absorbed and cause an uncontrolled temperature rise and subsequent fire. It is therefore of paramount importance to understand what these conditions are and to prevent them from occurring. Typically, abuse conditions can be categorized into three types, i.e. mechanical abuse (e.g. cell penetration), electrical abuse (e.g. overcharge) and thermal abuse (e.g. overheating). A common way to understand these conditions is through battery abuse testing. That is, batteries are subjected to non-nominal conditions in order to identify safety limits of the battery and worst-case scenarios. Appropriate controls can then be designed to minimize the risk of encountering such conditions during operations, and also related to quality control during manufacturing. Quality and manufacturing control is deemed out of scope of this discussion. Operational control could be to avoid unsafe conditions (external or internal) or to provide adequate cooling systems to be able to dissipate excess energy in case of an imminent thermal runaway event. Typically, the battery management system (BMS) monitors the cell voltage, current and possibly also the temperature to ensure proper control. This is an area of active research, and there are rapid developments in battery safety management [138]. It is noted in [139], that most fires and safety incidents with electrical vehicles (EV) have been due to one or more faulty cells reaching operating conditions beyond the safety limits. For larger systems, consisting of many cells, the risk of thermal runaway increases, and failures in one cell can propagate in a cascading manner to other cells in the battery system, as noted by [140]. It is reported in [123] that the rate of thermal runaway is higher in lithium-ion cells compared to sodium-ion and that therefore cascading heat propagation is more likely for lithium-ion systems.

Another aspect of maritime battery safety is the ability of the battery system to deliver the required energy and power at any time during operation. Failure to do so might cause loss of propulsion and maneuverability which might again lead to serious accidents such as collision and grounding. This safety aspect is typically monitored by combining state of charge and state of health of the battery. These two types of safety issues are referred to as thermal issues and functional issues, respectively, in [139].

A third aspect of battery safety is related to battery degradation, which may affect both thermal and functional performance of the batteries. It is well known that batteries age over time, and that this aging can have different effects on the battery. One is that battery capacity reduces by cycling and calendar time. Hence, the battery's ability to deliver the required energy and power is reduced over time. Another effect is that the battery internal resistance and impedance might increase, and this may increase the risk of thermal runaway and fire. It is generally assumed that degradation in terms of capacity loss and impedance increase is highly correlated, but this may not necessarily be the case for all battery chemistries. Other degradation mechanisms such as lithium plating may also affect a battery's thermal safety over time. The relationship between changes in thermal safety and electrochemical characteristics under different degradation paths is investigated in [141], and a method to evaluate thermal safety levels based on electrochemical observation indicators is proposed. At any rate, state of health is a measure of how much the battery has degraded, and different definitions of this metric exist; based on capacity and/or internal resistance.

For the first aspect of battery safety, related to fire safety, state of safety might be an interesting metric to monitor in a condition monitoring system. It has been suggested that ultrasonic monitoring can be used to detect physical changes in batteries and provide a means for early warning of thermal runaway without relying on the electrical measurements [142,143]. According to [142] it is possible to detect the start of overcharge in order to trigger an emergency stop and take the battery out of service before a catastrophic accident based on ultrasonic monitoring. However, only overcharge was investigated, and there is a need to extend to other failure mechanisms. A recent review of ultrasound sensing for lithium-ion state monitoring and thermal runaway detection is presented in [143], suggesting that ultrasonic monitoring allows for in-situ, non-destructive detection of internal failure mechanisms such as gas generation, swelling, lithium plating and aging in cells much earlier than with more conventional BMS systems. However, more research is needed for such systems to be used under real world operating conditions. A comprehensive review of early warning methods for thermal runaway of lithium-ion batteries is presented in [144], including early warning based on monitoring of internal electrical characteristics, temperature, force or gas generation, as well as ultrasound detection.

Alternative approaches to monitor and classify thermal safety status of batteries are explored in e.g. [145–147]. A thermal runaway warning method based on heat transfer theory and a simplified lumped parameter heat transfer model, as an alternative to more computationally demanding heat transfer finite element method, and a temperature distribution based on a few temperature sensors is presented in [145]. This simplified method is based on only three temperature sensors in a battery module and is intended to be included in the BMS in order to predict and calculate the temperature distribution within a battery module to identify faulty batteries before thermal runaway and hence provide an early warning for battery packs. Machine learning is used to develop an online safety classification method for lithium-ion batteries during charging and discharging in [146]. Four representative safety levels are defined and the model only needs a short period of current and voltage signals as input. However, it is noted that capacity loss is not considered, and this is a limitation of the current model. A simulation study analyzing the safety state based on characteristics values of voltage, temperature and deformation presented in [147] suggests that deformation will reach its warning threshold in advance of the other characteristics. Therefore, deformation can be used for early detection and warning of thermal runaway. However, when thermal runaway is triggered by overcharge the earliest detection is based on voltage signals. The battery doctor is introduced in [148] which aims to extend battery health monitoring from only considering state of charge and state of health to consider the battery's state of X (SoX), where SoX is formulated as a uniform indicator of the overall health conditions for battery cell, modules, packs and even large-scale battery systems.

More data-driven approaches to early warning for thermal runaway in batteries are also suggested in the literature. Warning methods for overcharge-thermal runaway for lithium-ion batteries based on features extracted from measurements of voltage, temperature and pressure is proposed in [149], based on association rules between features and thermal runaway warning levels. A multiphysics-informed deep learning model is proposed in [150] for predicting thermal runaway for lithium-ion batteries under thermal abuse. Deep learning is also used for thermal runaway warning in [151], based on coupled voltage–temperature information. The above references are merely examples, and there exist a huge amount of literature on early warning systems for thermal runaway of batteries. However, the focus on this literature is on the state of safety metric for monitoring the safety of batteries.

An SoS metric could integrate both aspects of battery systems into one, monitoring both conditions that might increase the risk of fire events and accounting for the effect of the gradual degradation. This is suggested in [139] as an important direction for future research. However, as discussed in [57] this may be very challenging, since two identical cells might respond differently to identical stress factors suggesting that probabilistic approaches are needed. Moreover, secondary batteries are complex highly nonlinear systems. Thus, large amounts of training data are needed, especially for training models to detect battery failures, since such failures are relatively rare. In the following, a brief literature survey on the notion of SoS will be presented.

The concept of state of safety was introduced in [152], and even though this paper is cited several times in the literature, only one additional reference, i.e. the master thesis presented in [153] has been found that really focuses on the SoS of batteries. Hence, the state of safety concept does not seem to have been widely implemented or further developed. In particular, no applications of the concept of state of safety have been identified for maritime battery systems. However, some related concepts, such as the overcharge safety status (OCSS) proposed in [154] and the safety risk function in [155], have been introduced.

According to [152], the state of safety can be defined as reciprocal to the concept of abuse in order to provide a quantitative measure of the safety of batteries. It is defined as a numeric quantity taking values from 0 (completely unsafe) to 1 (completely safe), in line with other state measures such as state of charge and state of health. The state of safety function is composed of sub-functions corresponding to various forms of abuse, e.g. related to voltage, temperature or mechanical deformation. The formulation is open for additional sub-functions to be added in order to generalize the concept of state of safety to cover other abuse conditions. Mathematically, the state of safety function is defined as follows:

$$f_{safety}(\boldsymbol{x}) = \frac{1}{f_{abuse}(\boldsymbol{x})} = \frac{1}{g(\boldsymbol{x})+1} \tag{1}$$

where $f_{abuse}(\boldsymbol{x})$ denotes the state of abuse, $f_{safety}(\boldsymbol{x})$ is the state of safety and $\boldsymbol{x}$ represents state and control variables describing the battery at a given time t. Examples of such variables are voltage, temperature, current, internal impedance, battery expansion and battery deformation [152]. The abuse function $g(\boldsymbol{x})$ is introduced to let state of safety vary between 0 for infinite abuse to 1 for zero abuse. A quadratic representation of the abuse function is suggested, to arrive at the following form

$$f_{safety}(\boldsymbol{x}) = \frac{1}{\left(\frac{1}{\zeta}-1\right)\left[\frac{h(\boldsymbol{x})-h(\boldsymbol{x}_{100})}{h(\boldsymbol{x}_{\zeta}-h(\boldsymbol{x}_{100}))}\right]^2+1}, \tag{2}$$

where $\boldsymbol{x}_{100}$ represents the combination of battery variables corresponding to 100% safety and $\boldsymbol{x}_{\zeta}$ corresponds to the variables describing the limit of acceptable safety, i.e. $f_{safety}(\boldsymbol{x}_{100}) = 1$ and $f_{safety}(\boldsymbol{x}_{\zeta}) = \zeta/100$, see [152] for details. It is observed that Eq. (2) takes the form of a Cauchy distribution. It is further suggested that such state of safety can be formulated for individual variables, using sub-functions, so that the overall state of safety can be calculated as the product distribution of the individual sub-functions, i.e.,

$$SoS(\boldsymbol{x}) = \prod_{k=1}^{n} f_k(x_n) \tag{3}$$

where f_k and x_k, $k = 1,\dots,n$ are the n individual state of safety functions and corresponding variables, respectively. Given the same limit ζ for safe behavior of each sub-function, then one would have some important values

SoS $= 1.0$: completely safe
SoS $= \zeta$: warning (one sub-function may be at ζ)
SoS $= \zeta^n$: minimum safety (all sub-functions might be ζ)
SoS $< \zeta^n$: unsafe (all sub-functions might be below ζ)

that can be used to trigger warnings and alarms. Note however, that it could be possible with this construction that all but one sub-functions are equal to 1.0 even if $SoS < \zeta^n$, indicating an unsafe condition in only one of the variables, so how to use the SoS as health indicators does not seem completely obvious. Note that similar safety functions were defined in [155], although with an overall score equal to the average of the individual safety scores. Moreover, [155] suggests adaptive fault thresholds based on confidence intervals derived from Gaussian processes regression.

Some of the variables relevant for state of safety that are discussed in [152] are temperature (over- and under-temperature), current (overcurrent), voltage (over- and under-voltage), state of charge (higher

SoC means more energy stored in the battery and hence less safety) and state of health. With regards to the latter, is suggested that SoH may have two opposite effects on safety; lower SoH means less energy stored even at maximum SoC and hence less fire potential and that an older battery (lower SOH) might be more prone to unsafe behavior and possible be damaged. It is hence possible to define two separate sub-functions for this variable in order to account for both effects. Other factors are internal impedance, mechanical deformation, time derivatives of any of the variables and combined conditions from dependent variables. It is further advised to establish safety limits by experimental tests for each of the relevant variables, and this method requires that the actual state of safety functions are defined for the battery system in question. It is noted in [156] that the fidelity and practicability of such an SoS implementation is difficult to guarantee, and that the accuracy will depend on the selected sub-functions.

This definition of state of safety is further investigated in [153]. The variables current, rack voltage, maximum and minimum cell voltage, cell imbalance and maximum and minimum cell temperature were used, and state of safety was monitored based on the following nine safety sub-functions: overcurrent, undercurrent, rack over- and undervoltage, cell over- and undervoltage, cell imbalance, overheat and overcold. Specific values for $\boldsymbol{x}_{100}$ and $\boldsymbol{x}_{\zeta}$ are assumed for each of the sub-function based on observations from normal operation. A large stationary storage system was monitored on rack level,[2] and data covering a period of 494 days from 198 racks were analyzed. Hence, data corresponding to 97 812 days of operation – nearly 141 million minutes – were available for the analysis. The state of safety classifies operations into safe, warning and critical conditions, and results are reported in terms of time spent in each of the warning/critical conditions for each of the sub-functions. It is reported that one or more abuses are identified for about 13% of the days (12 955 out of 97 182 rack days). Critical abuses are identified much less frequent than warning conditions, which is to be expected, except for cell imbalance. It is noted that although the study demonstrated that the concept of SoS is able to identify and classify warning and critical abuse states from operational data no actual thermal runaway event occurred during the data period, and proper validation of the results is difficult. Moreover, the study only looked at the various safety sub-functions separately, and did not aggregate them in an overall measure of state of safety as suggested in [152]. Another shortcoming of this application is that no sub-function was introduced to relate the state of safety to battery degradation and aging. Hence, an old battery would be regarded at 100% state of safety after experiencing abuse if the corresponding variables return within operational limits. It seems counterintuitive to allow the state of safety to return back to 100% when it is approaching end of life, so adding such a sub-function is believed to be important to obtain a more realistic measure for state of safety — it should reflect that batteries generally become less safe as they age [157].

In summary, the only definition of state of safety that is found in current scientific literature is the definition presented in [152]. Although this definition is referred to in several other papers, the authors of this paper have only found one additional reference that investigates this metric in detail, i.e. the master thesis in [153]. This study indicates that the state of safety, as defined in [152] is a feasible measure that can be used to monitor the safety level of a battery with respect to thermal runaway and fire. However, the various safety sub-functions and the various threshold levels need to be determined and further research is recommended to further develop and validate the SoS as a reliable measure for safety monitoring. It is believed to be important to include a sub-function that can account for the effect of aging and degradation on the safety of batteries. This can be included in the overall definition of SoS, although it was not implemented in the SoS measure investigated in [153]. Moreover, it is assumed that the correlations and interdependencies between the input variables and sub-functions might be important so a more robust system should account for these. In addition, it is believed to be interesting to investigate how to build in some kind of memory into the state of safety measure, assuming that a battery that has experienced abuse might be less safe than an equally old battery (with the same SoH) that has never experienced abuse. It is not obvious how this should best be incorporated in a state of safety measure, and more research is recommended to further develop these aspects of battery state of safety.

Based on the above discussions, it is believed that the state of safety (SoS) has great potentials to ensure safe and reliable energy storage solutions during the design, operation, and maintenance phases in maritime battery systems. It involves real-time monitoring and management of critical safety parameters such as temperature, voltage, current, and state of health, as well as predictive diagnostics to prevent hazardous conditions like thermal runaway or electrical faults. However, the effective implementation of SoS also faces several obstacles such as the lack of standardized safety protocols across diverse battery technologies, the complexity of integrating state-of-safety assessments with existing ship systems, and the high cost of advanced monitoring and control systems. Addressing these challenges requires industry-wide collaboration, regulatory support, and advancements in robust and cost-effective safety technologies.

Conclusions

Implementing State of Health (SoH) and State of Safety (SoS) monitoring in real maritime applications presents several challenges across technical, economic, and operational dimensions. Technically, maritime environments impose demanding conditions such as vibrations, extreme temperatures, and humidity, which can compromise the reliability of sensors and monitoring systems. Accurate SoH and SoS assessments require advanced algorithms capable of processing large volumes of data in real-time, which may be constrained by limited onboard computational resources and communication bandwidth. Additionally, the diverse chemistries and configurations of maritime battery systems complicate the standardization of monitoring frameworks. Economically, the high initial investment in advanced sensors, data analytics platforms, and integration with existing ship systems can be a barrier, particularly for smaller operators. Furthermore, ongoing costs for calibration, maintenance, and potential retrofitting add to the financial burden. Operationally, integrating SoH and SoS monitoring into the workflows of maritime personnel requires adequate training and user-friendly interfaces to avoid misinterpretation of critical safety data. Ensuring compatibility with regulatory requirements and addressing cybersecurity risks associated with interconnected monitoring systems are further operational hurdles. Overcoming these challenges demands collaborative efforts among stakeholders to develop cost-effective, robust, and standardized solutions tailored to the maritime context.

This paper has presented a literature review on data-driven online state of health estimation for alternative battery chemistries, and on state of safety monitoring. The overall idea has been to see what type of approaches have been suggested for SoH prediction for LFP-type lithium-ion and sodium -ion batteries, and how these differ from methods applied to NMC type of lithium ion batteries. Based on the literature review, there are no reason to believe that data-driven approaches to state of health estimation should be fundamentally different for these alternative chemistries compared to NMC-type of batteries. Hence, approaches such as empirical and semi-empirical methods, physics-based models and purely data-driven machine learning techniques are believed to be applicable across chemistries types. However, it is emphasized that different battery types are expected to experience different degradation paths and respond differently to various loading conditions so it is important to establish battery-specific models. Hence, model training, calibration, fine-tuning and validation needs to be

[2] For this system, a rack is equivalent to a string

performed on a case-by-case manner. Notwithstanding, one particular feature of LFP-type lithium-ion batteries should be acknowledged; they are typically associated with very flat OCV curves. This means that methods relying on OCV might perform poorly on LFP-type of batteries. This is not the case for sodium-ion batteries, which are associated with comparable large voltage drops during discharge.

The concept of a measure for the state of safety of batteries is relatively novel and is not yet well established in the industry. The most common definition is that the state of safety can be regarded as the reciprocal of the state of abuse, and various safety sub-functions can be defined based on different abuse conditions, e.g. thermal, voltage and mechanical abuses. In addition, safety sub-functions can be defined to account for the effect of battery aging on the thermal safety, e.g. related to SoH. These safety sub-functions are functions of various state and control variables that should be collected and used as input for calculating SoS. Although there does not seem to be much experience with the state of safety, and no experience at all in maritime applications, this measure seems highly relevant for maritime battery systems. If successfully implemented, it is believed that it may provide a means for monitoring when the battery experiences abusive conditions and hence provide a means for early warning of impeding safety events and possible thermal runaway. This is deemed particularly important for very large onboard battery systems, with large fire potential. Nevertheless, the maturity of this measure is low, and further investigations are recommended to explore optimal safety sub-functions and associated warning thresholds. Such a system needs to be carefully calibrated and fine-tuned to ensure it can reliably predict critical conditions without frequent false alarms. Moreover, it needs to be investigated how such a system scales to very large battery systems. It also needs to be considered how the state of safety can be exploited from a classification point of view to enhance the safety of electric ships.

CRediT authorship contribution statement

Erik Vanem: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Formal analysis, Conceptualization. **Shuai Wang:** Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Erik Vanem reports financial support was provided by Research Council of Norway. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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